

MA378: Tutorial Sheet 5

All questions are from old exam papers; some likely featured on other Tutorial Sheets

Also: Let p_n be the polynomial of degree n that interpolates f at the $n + 1$ points $a = x_0 < x_1 < \dots < x_n = b$. You may assume that, for any $x \in [a, b]$ there is a $\tau \in (a, b)$ such that

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\tau)}{(n+1)!} \pi_{n+1}(x), \quad (1)$$

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Q1. (Q2, 24/25)

- (a) Let $f(x) = xe^x$. Calculate the Lagrange form of p_2 , the polynomial interpolant to f at the interpolation points $\{x_0, x_1, x_2\} = \{0, 1, 2\}$.
- (b) Using Equation (1), find an upper bound for $|f(x) - p_2(x)|$, at $x = 1.5$.
- (c) Let p_n be the polynomial of degree n that interpolates an arbitrary function f at the $n + 1$ equidistant points $-1 = x_0 < x_1 < \dots < x_n = 1$.
Can one expect that $\max_{-1 \leq x \leq 1} |f(x) - p_n(x)| \rightarrow 0$ as $n \rightarrow \infty$? Explain your answer.

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Q2. (Q3, 24/25)

- (a) Define the *natural piecewise cubic spline interpolant* to the function f that is continuous on $[a, b]$ at the $N + 1$ knot points $\{x_0, x_1, \dots, x_N\}$.
- (b) Suppose that S is a natural cubic spline on $[0, 2]$ with

$$S(x) = \begin{cases} 10x + a(1-x)^2 + x^3, & \text{for } 0 \leq x < 1 \\ b(2-x) - c(x+1)^2 + d(x-1)^3, & \text{for } 1 \leq x \leq 2. \end{cases}$$

Find a, b, c and d .

- (c) Suppose that S is the cubic spline interpolant to $f(x) = \ln(x)$ on the $N + 1$ equally spaced points $\{x_0 = 1 < x_1 < \dots < x_N = 4\}$. The error estimate between the function f and the cubic spline interpolant S has an upper bound given by

$$\|f - S\|_\infty := \max_{x_0 \leq x \leq x_N} |f - S| \leq \frac{5h^4}{384} \max_{x_0 \leq x \leq x_N} |f^{(4)}(x)|,$$

where $h = \frac{x_N - x_0}{N}$.

What value of N should one take to ensure that $\|f - S\|_\infty$ is no more than 10^{-6} ?

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Q3. (Q4, 24/25) Consider the following differential equation:

$$-u''(x) + u(x) = x \quad \text{on } (0, 1) \quad \text{with } u(0) = u(1) = 0 \quad (2)$$

- (a) State the variational formulation of the differential equation (2).
Show that the solution to the variational problem is unique.
- (b) Suppose you wanted to compute an approximation to (2) using the finite element method (FEM) on the uniform mesh $\{x_0, x_1, \dots, x_n\}$. Denote the set of the usual piecewise linear Galerkin basis ("hat") functions as $\{\psi_1, \psi_2, \dots, \psi_n\}$.

- i. Give a formula for a typical basis function ψ_i and give a rough sketch of the basis function.
 - ii. The FEM applied to (2) leads to a linear system of equations, which can be written as the matrix-vector equation $Ax = b$. Give an expression for the entries of A and b in terms of the basis functions ψ_i . (You do not have to derive an explicit formula for these entries.)
 - iii. Explain why A is symmetric and tridiagonal.
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Q4. (Q2, 23/24)

- (a) Let $\{x_0, x_1, x_2\} := \{1, 4, 9\}$ be a set of interpolation points. Write down the formulae for the associated Lagrange Polynomials, $\{L_0, L_1, L_2\}$, and give a rough sketch of them, clearly indicating their values at the interpolation points. (You can make the sketch in your answer book; graph paper is not needed).

Suppose we define $q(x) := 1 - L_0(x) + L_1(x) + L_2(x)$. Show that, in fact, $q(x) \equiv 0$, for all x .

- (b) Let $f(x) = x^{3/2}$. Give the Lagrange form of p_2 , the polynomial interpolant to f at $x_0 = 1$, $x_1 = 4$, and $x_2 = 9$.

Using (1), give an upper bound for $|f(7) - p_2(7)|$.

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Q5. (Q3 23/24)

- (a) Let l be the piecewise linear interpolant to a given function f on $N + 1$ *equally spaced points* in the interval $[a, b]$. Let $h = (b - a)/N$. Use (1) to give an upper bound for $\max_{a \leq x \leq b} |f(x) - l(x)|$, in terms of h and f'' .

- (b) Let $f(x) = \ln(x)$, $a = 1$, $b = 3$ and $N = 2$. Write down a formula for its piecewise linear interpolant l .

Give an upper bound for $\max_{a \leq x \leq b} |f(x) - l(x)|$.

What value of N would you have to choose to ensure that $\max_{a \leq x \leq b} |f(x) - l(x)| \leq 10^{-4}$?

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Q6. (Q4, 23/24)

- (a) We denote the Newton-Cotes quadrature rule for approximating $\int_{-1}^1 f(x)dx$ as

$$Q_N(f) := \sum_{i=0}^N q_i f(x_i).$$

Show that $q_i = q_{N-i}$ for $i = 0, 1, \dots, N$.

- (b) Consider the following rule for approximating $\int_0^4 f(x)dx$:

$$R(f) = q_0 f(0) + q_1 f(1) + \frac{2}{9} f(3) + f(4).$$

- i. Determine the values of q_0 and q_1 so that $R(\cdot)$ has precision 1.
- ii. What is the maximum precision of $R(\cdot)$ for the values of q_0 and q_1 that you have determined?
- iii. Why is $R(\cdot)$ not a Newton-Cotes rule, in the strictest sense?